

WEEK-4 PROJECT – Student Performance Prediction Summary Report Final Summary

Key Findings

- Analysis of 10,000 student records revealed that Previous Scores is the strongest predictor of academic performance, with a correlation of 0.91 confirming that past performance is the most reliable indicator of future results.
- Hours Studied ranked second with a coefficient of 2.85, meaning every additional study hour adds nearly 3 marks to the predicted score.
- Tuition attendance showed the highest coefficient overall (2.99), demonstrating that structured coaching provides a measurable advantage.
- Female students scored approximately 2.5 marks higher on average than male students across all study levels.
- Extracurricular Activities and Sample Papers Practiced had minimal impact, with coefficients of 0.63 and 0.19 respectively.

Conclusion

- The Linear Regression model achieved an RMSE of 2.11 marks and an R^2 Score of 0.9875, with 98.2% of predictions falling within ± 5 marks of the actual score.
- These results confirm that student performance is largely driven by prior academic history and consistent study effort both measurable and actionable factors.
- Schools can leverage such predictive models to identify at-risk students early and design targeted interventions, while students themselves can use these insights to prioritise the habits that matter most.